

CIT 3302 ADVANCED DATABASE SYSTEMS

Pre-requisite CIT 3201 -Database Systems

Purpose of the Course

The course is designed to equip the student with **advanced** knowledge of database systems in terms of design, implementation, and proficiency in data management skills.

Expected Learning Outcomes

By the end of this course the student should be able to:

- Understand the fundamental concepts, structures, operations and functions of different components of Databases.
- Describe the roles of transaction processing and concurrency control in a modern DBMS.
- Understand the issues involved in the operation of a DBMS including query processing, security and integrity.

Course Content

Transaction Management: The Concept of a Transaction, Transactions and Schedules, Concurrent Execution of Transactions; Concurrency Control: Lock-Based Concurrency Control, Lock Management, Specialized Locking Techniques, Concurrency Control without Locking; Query Processing: Details of the conceptual and logical database, design procedure for an enterprise level database; Query Optimization, Relational Algebra ; Database Security and Recovery, Distributed Databases, Decision Support Systems; Object Database Systems: Object Oriented Database Systems, Object Relational Database Systems, Database Design for an ORDBMS, Comparing OODBMS and ORDBMS.

Mode of Delivery

Lectures, Discussions, Presentations, Quizzes, Independent Study

Instructional Materials and Equipment

Computers, E-content, DBMS & Tools

Course Assessment

Continuous Assessment	30%
Examination	70%
Total	100%

Core Reading Materials

1. Coronel et al (2010) Database systems – Design Implementation and management (9th Ed.) Cengage Learning.
2. Connolly, T. M., & Begg, C. E. (2005). Database systems: a practical approach to design, implementation, and management. Pearson Education.

Week	Topic	Sub Topics
Week 1	Query Processing	– Recap of a relational database model – What is query processing – Steps involved in query processing – Basic query operations
Week 2	Relational Algebra	– Presentation of basic query operations – Presentation on queries with multiple tables – Equivalence rules
Week 3	Query Optimization	– Importance of query optimization – Basic Steps in query optimization – Cost-based optimization – Rule-based optimization
Week 4	Transaction Processing	– What is a database transaction? – Transactions states – Transaction Management – Serial scheduling
Week 5	Concurrency Control	– What is concurrency control? – Why is concurrency control necessary? – Concurrency control protocols – Deadlocks and starvation
Week 6	Database Security	– What is database security? – Classification of database security – Database access controls
Week 7	Database Recovery	– What is database recovery? – Types of database failures – Database recovery facilities
Week 8	CAT	
Week 9	Database fragmentation	– What is database fragmentation? – Objectives of database fragmentation – Issues in database fragmentation – Types of database fragmentation
Week 10	Database replication	– What is database replication? – Objectives of database replication – Types of database replication – Replication schemes
Week 11	Distributed Databases	– Distributed vs. centralized databases – Features of distributed databases – Objectives of database distribution – Challenges of data distribution
Week 12	Distributed Database Architectures	– Client-server architecture in distributed architecture – Collaborating server architecture – Middleware architecture
Week 13	NoSQL and NewSQL Databases	– Benefits of NoSQL databases – Challenges of NoSQL databases – Applications of NoSQL databases
Week 14	Revision	
Week 15	Revision	